



HAWTHORN RESIDENCE - JANE RIDDELL ARCHITECTS

MASTERWALL

The MasterWall Insulated External Wall System is designed to provide a long-lasting, energy efficient, weather-tight building envelope with architectural integrity and excellent performance credentials.

Masterwall offers designers the ability to create without compromise and builders to construct quickly, effectively and economically whilst reducing the carbon footprint of the building. That's why MasterWall is the leading choice of Australian home owners, builders, architects and designers.

SHARING OUR RESEARCH AND KNOWLEDGE

When you choose MasterWall EIFS you not only get Australia's leading Insulated Wall System, you also are assured industry-leading support and the reassurance that all our Distributors and Accredited Installers are trained experts in the MasterWall System:

- Nationwide telephone technical support.
- Online resources including detailed construction manuals.
- Regular building seminars across metropolitan and regional Australia.
- Regular feedback to regulatory bodies; building surveyors (AIBS), Councils, the CSIRO & the MBA.

NATIONAL DISTRIBUTION

MasterWall is available in most parts of Australia via our network of capital city and regional distributors who are experts in delivering the MasterWall System.

WARRANTY

PracticaMMC Pty Ltd warrants that its products are free from defects in materials and workmanship for a period of 7 years from the date of purchase. (For a full description of the Warranty refer to the MasterWall Product Specification and Installation Manual).

ONLINE SUPPORT

Further details are available on the MasterWall website including downloadable brochures:

1.



Step By Step Guide

2.



Product Specification & Installation

3.



Construction Details

All of these booklets are available online at:
www.masterwall.com.au

YOUR LOCAL ACCREDITED INSTALLER / DISTRIBUTOR:

TO LEARN MORE:
WWW.MASTERWALL.COM.AU

OR CALL US DIRECT:
(03) 9553 3211

ADDRESS: 9-11 Wren Road, Moorabbin 3189, Victoria, Australia
PO BOX: 3168 Moorabbin East, Victoria 3189
EMAIL: sales@practicammc.com.au **FAX:** 03 8678 3238



MASTERWALL

MASTERFLOOR

MASTERWALL
K-SERIES



PracticaMMC is an Australian company offering a unique range of modern, insulating wall and floor construction solutions for residential & commercial projects. www.practicammc.com.au

ABOUT MASTERWALL

The MasterWall Insulated External Wall System has been specifically developed to meet Australian conditions and is based on proven technology used extensively throughout Europe and North America for more than 60 years.

No other cladding product can claim such a wide range of product benefits including - Energy efficiency, Design flexibility and Economy.

Studies in the U S have shown that Insulated External Wall Systems are the best performing wall systems and, "out performed all other walls in terms of moisture while maintaining superior thermal performance"

Source: Oak Ridge National Laboratory Research

THE SYSTEM

The MasterWall System is an external lightweight, reinforced, insulating polystyrene wall panel, mechanically fixed to the outer face of the building whether that be timber or steel framing or a masonry substrate. The complete system includes the application of approved acrylic render systems, frame wraps, trims, sealants and opening flashings at openings that combine to create a weather-tight building envelope.

Panel: The base material is Medium (M) Grade expanded polystyrene (EPS) in sheet form with an added fire retardant.

Facing: Flexible cementitious-based material designed for both an excellent mechanical fix and a natural chemical bond to the polystyrene cladding, using the binders and polymers found in approved acrylic render systems. The finished surface of the panel has a light trowel type texture.

Reinforcing: This facing is reinforced with an alkaline-resistant fiberglass mesh, 145gsm/m² minimum and is compatible with approved acrylic render and decorative finishes.

Dimensions:

Standard panel size:	2400 x 1200mm
Nominal thickness:	50, 75, 100, 125mm
Area:	2.88m ²

Mass:

Thickness	50mm	75mm	100mm	125mm
kg/m ²	2.4	2.9	3.4	3.9
Total sheet kg	6.9	8.4	9.8	11.4

PERFORMANCE

Thermal Performance - R Rating Values:

Thickness	Rating	R ^t
125mm	3.4	3.7
100mm	2.7	3.0
75mm	2.0	2.3
50mm	1.3	1.6

NOTE: R^t refers to the total thermal value of a wall system which includes an allowance of R 0.3 that can be attributed to a standard stud frame without batts and an internal lining of 10mm plasterboard.

Fire Performance

Comparative testing of some materials to AS1530, Part 3 — Early Fire Hazard Test.

Material	Ignitability (0-20)	Spread of Flame (0-10)	Heat Evolved (0-10)	Smoke Developed (0-10)
Polystyrene	12	0	3	5
Softboard	16	9	7	3
Oregon	13	6	5	3
Bluegum	11	0	3	2

Source: EBS Notes on Science of Building NSB66

Wind load performance:

As a testament to its strength, the MasterWall System has achieved a **Cyclonic Rating (C3)**, the toughest test of an external wall system in Australia. Under test conditions Masterwall achieved windloading of 4.1kpa to the AS-40403.

Weather-tight performance:

The MasterWall System includes the installation of window flashings, and joint and penetration sealant processes that ensure a weather-tight building envelope, even before the application of the render system. To achieve the best performance, thermal bridging, weather tightness (and therefore air tightness), and the potential for moisture penetration must be addressed when considering the design of the external envelope of a building.

Other performance properties:

For a full description of the MasterWall Insulated External Wall System please consult the MasterWall Product Specification and Installation Manual available in the Technical Centre on the website.
WWW.MASTERWALL.COM.AU

MASTERWALL

COMPARE THE BENEFITS OF THE MASTERWALL SYSTEM

ADD UP THE ADVANTAGES
OF MASTERWALL AND YOU'LL
DISCOVER A SYSTEM THAT
DELIVERS BETTER VALUE,
FASTER INSTALLATION
AND SUPERIOR QUALITY.



BCA
ACCREDITATION

MASTERWALL IS THE FIRST EXTERNAL WALL CLADDING SYSTEM IN AUSTRALIA TO DEMONSTRATE COMPLIANCE WITH THE PERFORMANCE REQUIREMENTS OF THE BUILDING CODE OF AUSTRALIA (BCA). BUILDING COMMISSION ACCREDITATION NO. V09/01.

BCA COMPLIANCE

All external wall systems utilised in Australia are required to meet the Performance Requirements as prescribed in the Building Code of Australia (BCA).

To meet the Performance Requirement of the BCA, the system supplier must provide an independent certificate of Accreditation or an Appraisal from one of the following Registered Testing Authorities (RTA):
The Building Commission (Accreditation), Codemark (Accreditation), BRANZ (Appraisal).

NOTE: The Building Commission and the Australian Building Codes Board (ABCB) have recently reiterated that Accreditations or Appraisals that relate to a product only such as white/raw polystyrene, rather than a product and its unique building system, do not meet the Performance Requirement for Alternative Solutions and therefore may not be used in the construction of a dwelling.

For copies of Advisory Notes from both the Building Commission and the ABCB go to the MasterWall website.

DURABILITY

Lightweight polystyrene construction has impressive impact resistance, including a high degree of compressive and flexural strength and dimensional stability, which provides opportunities for innovation in design and construction. MasterWall is the only system that can provide a completely weather-tight solution that eliminates water ingress and protects the structure from moisture damage. The C3 Cyclone Rating demonstrates its superiority under extreme weather conditions.

SAVE TIME AND MONEY

The lightweight construction speeds up installation ensuring minimum time to lock-up. Depending on the design, the MasterWall System can be more than 30% cheaper than rendered brick veneer.

REDUCE YOUR ENERGY COSTS

MasterWall gives you excellent thermal insulation, with the flexibility of a choice of R ratings, up to R 3.4, to suit individual applications. Traditional insulation in between studs has "thermal breaks", gaps where heat and cold pass more freely between the outdoors and space within. By wrapping the house in a seamless blanket it stabilizes the interior environment and reduces energy consumption.

DESIGN FREEDOM

A designer's dream, the MasterWall System offers the ability to design features such as curved walls and with different thicknesses, create light-weight second storey structures or apply different colours and textures all of which would be prohibitively expensive if using any other material.

TECHNICAL SUPPORT

PracticaMMC has an outstanding commitment to providing complete product support including an online Technical Centre - a one-stop resource for information about the MasterWall System. PracticaMMC also offers a nationwide telephone technical help line offering advice from building design and installation techniques to help ensure that the MasterWall System is installed to provide optimum performance of the external building fabric.

MASTERWALL.

THE FLEXIBILITY TO DESIGN GREAT SPACES

- LIGHT-WEIGHT CONSTRUCTION
- VARIETY OF TEXTURES & FINISHES
- CURVED FACADES OR CLEAN, STRAIGHT LINES
- FULLY WEATHER-TIGHT, MONOLITHIC FINISH ON WALLS, SOLID BALCONY BALUSTRADES AND PARAPETS
- BRICK-LIKE WINDOW AND DOOR REVEALS



ESTIMATING R VALUES*

Thermal rating of common wall construction materials

As owners and builders become more conscious of the rising financial and environmental costs of energy, thermal efficiency is becoming increasingly important. Government energy rating standards for residential homes is an example of the initiatives that are driving the way we build and renovate.

Whilst a building's overall energy rating incorporates many different factors, core considerations are the thermal efficiency of the walls, ceilings and floors. Insulation of the building envelope keeps heat in during winter, but also keeps heat out during summer to improve comfort and save energy. A well insulated home can save an average of 20% of energy costs per year. Whilst design can certainly affect thermal efficiency, the correct choice of wall cladding materials is the single most important consideration.

An R Rating of up to R 3.7 can be achieved with MasterWall's 125mm polystyrene panel, stud frame and 10mm internal plasterboard.

WALL CONSTRUCTION	R VALUE
MasterWall (125mm)	R 3.4
Weatherboard	R 0.55
Brick veneer	R 0.51
Solid brick (230 mm thick)	R 0.44
Solid concrete (100 mm thick)	R 0.23
Solid concrete (200 mm thick)	R 0.30
Aerated concrete (100 mm block)	R 0.78
Aerated concrete (200 mm block)	R 1.54

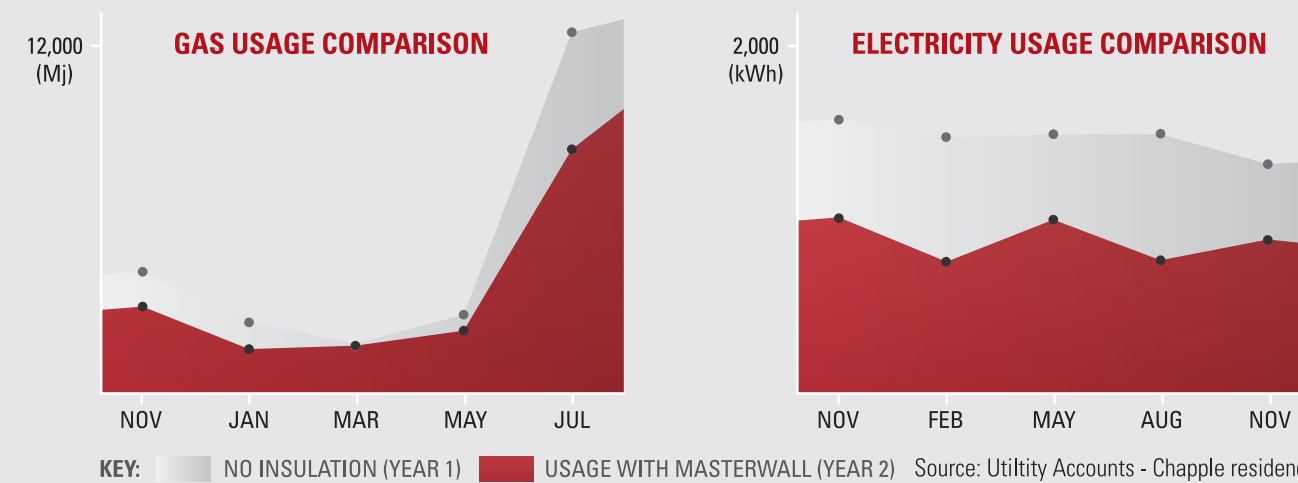
*As R value increases, the insulation benefit improves.
Source: Sustainable Energy Authority Victoria 2002, MasterWall (CSIRO tests)

HOUSEHOLD ENERGY BILLS - WITH AND WITHOUT MASTERWALL

Most people understand the long term value of installing solar panels on the roof of their home, but it's not necessarily the same case when it comes to insulating and weatherproofing. In winter, heat loss occurs through walls and ceilings that are poorly insulated, and gaps in floorboards and wall cavities is a major source of heat loss. To prove the point, MasterWall spent 12 months collaborating with one of its customers after installing the MasterWall System. Energy consumption pre and post installation was compared, with the results proving what the owners experienced day-to-day (see graph diagram below).

"Over the last 12 months I have notice a huge change in the consistency of the temp in the house when either the heating or cooling is on, the house remains warm in winter and cool in summer without the need for turning on the heater or airconditioner day and night".
~ Jo Chapple.

The results are even more dramatic given the household went from 2 adult occupants between Nov to May of the first year to 3 adults thereafter. The MasterWall System is the only system on the market that provides a total, weather-tight insulation solution whilst at the same time significantly increasing the external design-value of your home.



POLYURETHANE FOAM SEALANT:

100% flexibility ensures integrity and weatherproofing of joints.

MESHED & PRIMED POLYSTYRENE:

M grade polystyrene, light weight, designed to move with the frame
(Thicknesses: 50/75/100/125mm R 1.6/2.3/3.0/3.7).*

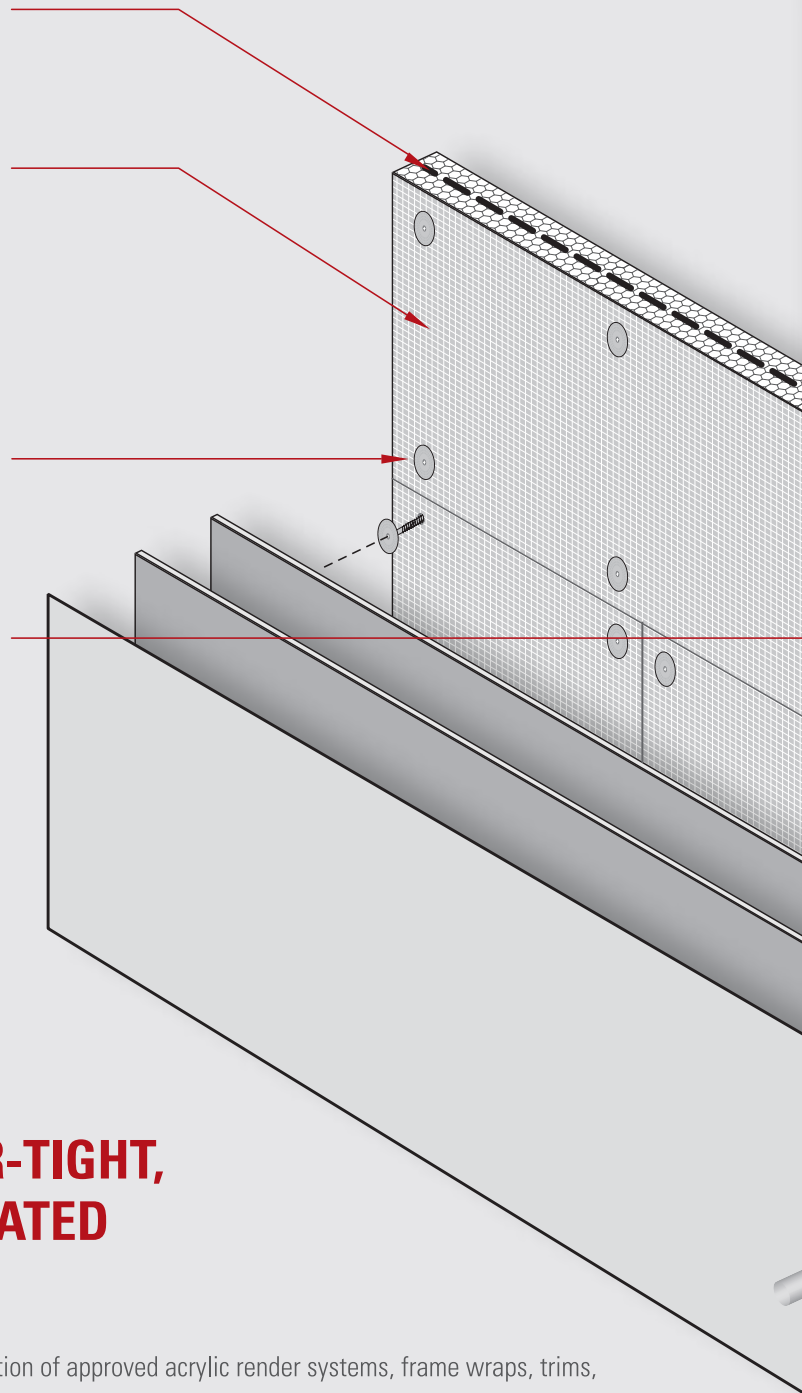
* Includes R 0.3 allowance for 90mm timber frame and 10mm plasterboard.

FIXINGS:

Pre-assembled 50mm button and class 3 screw sets provide the right fix for every situation.

ALLOY EXTERNAL CORNERS:

With integrated reinforcing mesh for strength.

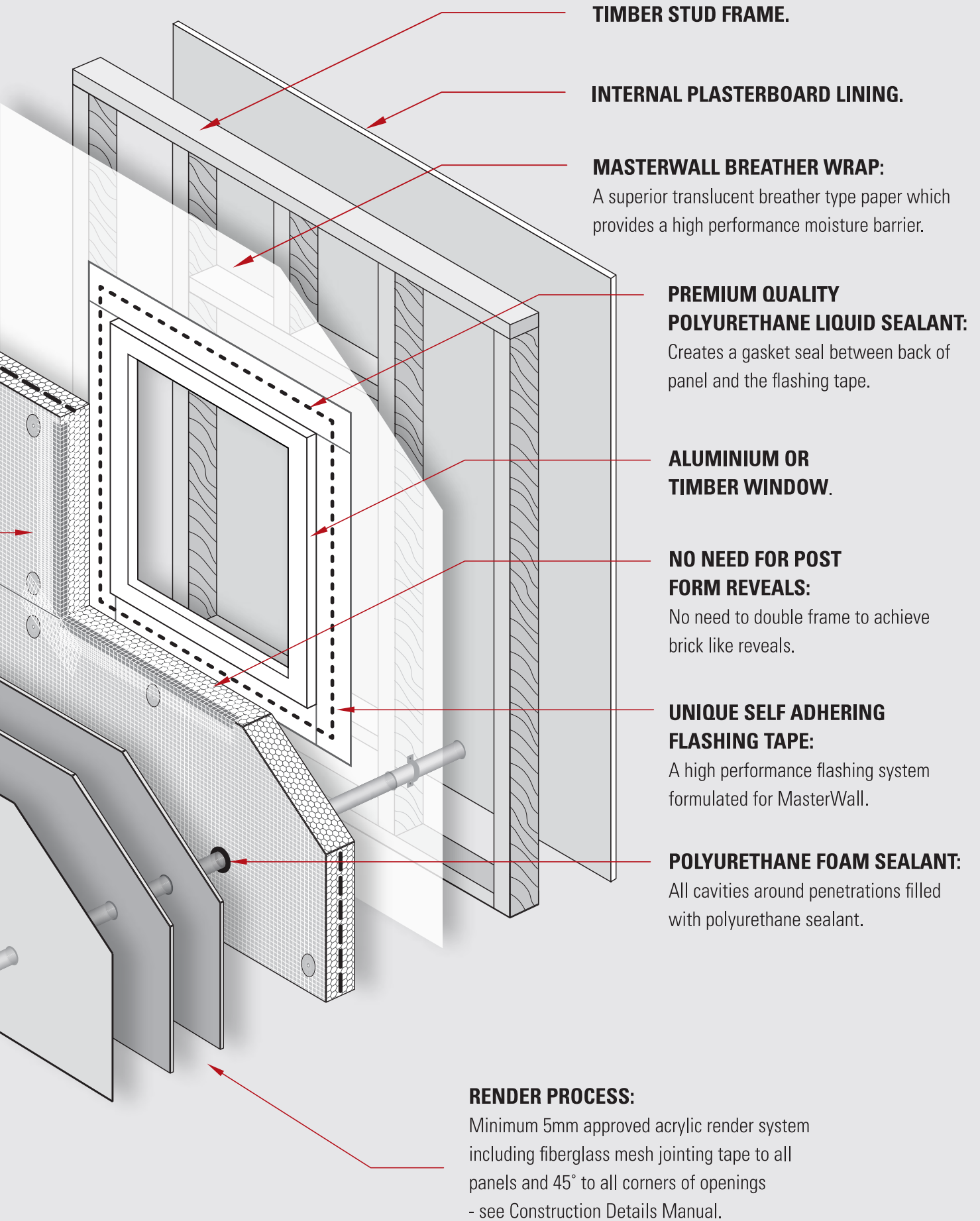


HOW WE BUILD A WEATHER-TIGHT, ENERGY EFFICIENT & INSULATED WALL SYSTEM

MasterWall is a complete system which includes the application of approved acrylic render systems, frame wraps, trims, sealants and opening flashings. When combined, these elements provide a complete seal between the exterior and interior of a building, thereby keeping the weather and moisture out and significantly raising the overall insulation performance when compared with stand-alone products such as glass wool batts.

Apart from giving you great insulation performance, MasterWall is also fire-resistant, achieves a C3 Cyclonic Rating and offers excellent acoustic performance.

From a design perspective, the system can be finished in a variety of architectural textures and finishes to create the look you want.



Note: Diagram not to scale.